

A PROJECT TITLE
ON
Movie Recommendation System Using Collaborative Filtering and Deep Learning
project report submitted in partial fulfilment of the requirements for the Course
MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

Name of Title: Movie Recommendation System Using Collaborative Filtering and Deep Learning

1. Abstract:

- The rapid growth of online streaming platforms has significantly increased the amount of digital entertainment content available to users, making it difficult to identify movies that match individual preferences. Recommendation systems have become an essential solution for improving user experience by providing personalized suggestions based on user interests and historical interactions. This project presents a Movie Recommendation System Using Collaborative Filtering and Deep Learning to generate accurate and personalized movie recommendations. The system is developed using the MovieLens Latest (ml-latest) dataset, which contains information about movies, user ratings, tags, and links. The dataset is preprocessed by handling missing values, removing duplicate records, encoding user and movie identifiers, and preparing the data for model training. Two recommendation approaches are implemented: a Collaborative Filtering model using Singular Value Decomposition (SVD) as the baseline and a Neural Collaborative Filtering (NCF) model developed using TensorFlow/Keras with embedding layers to learn complex user-movie relationships. The performance of both models is evaluated using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Mean Squared Error (MSE) to measure prediction accuracy and recommendation quality. The project also includes exploratory data analysis, feature engineering, model comparison, visualization of results, and a recommendation engine capable of generating personalized movie suggestions for users. The deep learning model is expected to capture nonlinear interactions between users and movies more effectively than the traditional collaborative filtering approach, thereby improving recommendation accuracy. The complete implementation is carried out using Google Colab, Python, TensorFlow/Keras, Scikit-learn, Surprise, Pandas, NumPy, and Matplotlib. The proposed recommendation system demonstrates the practical application of machine learning and deep learning techniques in solving real-world personalization problems and highlights how intelligent recommendation algorithms can enhance user satisfaction, increase content discovery, and support modern online entertainment platforms through efficient and accurate movie recommendations.

2. INTRODUCTION:

The rapid growth of online streaming platforms has led to an unprecedented increase in the availability of digital entertainment content. While this provides users with a wide variety of choices, it also creates the challenge of information overload, making it difficult to identify movies that match individual preferences. Traditional search methods and generic recommendations often fail to capture the unique interests of each user, resulting in lower user satisfaction and reduced content discovery. Recommendation systems have therefore become an essential component of modern streaming platforms, as they analyze user behavior and historical interactions to generate personalized movie suggestions. By delivering relevant recommendations, these systems enhance user experience, reduce search time, and improve engagement with digital entertainment services.

The motivation behind this project is to develop an intelligent and accurate Movie Recommendation System Using Collaborative Filtering and Deep Learning that addresses the limitations of conventional recommendation techniques. Collaborative Filtering utilizes user rating patterns to identify similar preferences, while Deep Learning models, particularly Neural Collaborative Filtering (NCF), are capable of learning complex and nonlinear relationships between users and movies. By combining these approaches, the proposed system aims to generate highly personalized and reliable movie recommendations. The project demonstrates the practical application of machine learning and deep learning techniques in solving real-world recommendation problems, ultimately enhancing content discovery, increasing user satisfaction, and supporting the evolving needs of modern online entertainment platforms.

3. SOFTWARE REQUIREMENTS:

The proposed Movie Recommendation System is developed using Google Colab as the development environment with Python as the programming language. The project utilizes machine learning and deep learning libraries such as TensorFlow/Keras, Scikit-learn, Surprise, Pandas, NumPy, and Matplotlib for data preprocessing, model training, evaluation, and visualization. The MovieLens Latest (ml-latest) dataset is used to train and evaluate the recommendation models. A computer with standard hardware specifications and a stable internet connection is sufficient to develop, train, and test the proposed system.

Category	Requirement
Operating System	Windows 10/11, Linux, or macOS
Development Environment	Google Collab
Programming Language	Python 3.x
Dataset	MovieLens Latest (ml-latest)
Machine Learning Libraries	Scikit-learn, Surprise
Deep Learning Framework	TensorFlow / Keras
Data Processing Libraries	Pandas, NumPy
Visualization Library	Matplotlib
Evaluation Metrics	RMSE, MAE, MSE
Processor	Intel Core i5 or equivalent
RAM	Minimum 8 GB
Storage	Minimum 2 GB free space
Internet	Required

*Signature of the Guide

Course Instructor